Implementation of FastSpeech 2-Based Text-to-Speech Synthesis

English (Svarah) and Telugu (IndicTTS) · Mahitha Chopra Lankapalli · Technical Report

1. Summary

Two separate FastSpeech 2 models were trained from scratch (no pretrained acoustic-model weights): English on Svarah and Telugu on IndicTTS Telugu, each on about 4–5 h of training audio and about 1 h of validation audio. The HiFi-GAN "universal" vocoder from the reference repository is used unchanged. Ten test sentences (E1–E4, T1–T4, as specified) were synthesised and their predicted duration, pitch and energy plotted.

Honest outcome: the pipeline works end to end (alignment, training, inference, plots) and the vocoder is not the bottleneck (re-synthesising ground-truth mel-spectrograms with it gives clean audio). However, the acoustic models trained on this little data **overfit early and the synthesised speech is mostly not intelligible**. Only the first one to two seconds of some clips (Telugu T2a/T2b) were judged understandable by listening; the English clips were noisy to unintelligible. The sections below document what was done, what failed, and why.

2. Implementation and modifications

Base code: ming024/FastSpeech2 (LibriTTS configuration): 4-layer FFT encoder, 6-layer FFT decoder, 2 heads, hidden size 256, conv filter 1024 (kernel 9/1), dropout 0.2; variance adaptor with duration, pitch and energy predictors (filter 256, kernel 3, dropout 0.5), phoneme-level pitch and energy with 256 linear quantisation bins, `max_seq_len` 1000. Mel target 80 bins, 22 050 Hz.

Area	Change
Library compatibility	librosa 0.11 keyword arguments in `audio/stft.py` / `audio_processing.py`; energy targets cast to float32 (float64 broke back-propagation); `torch.load(..., weights_only=False)` for PyTorch 2.6+.
Train/validation split	Preprocessor reads the split from fixed manifest files instead of a random shuffle, so the split is reproducible and has no overlap.
English	Multi-speaker (115 speaker embeddings). MFA pretrained `english_us_arpa` alignment. Frontend: text normalisation (times, years, numbers to words, `en_norm.py`), CMU lexicon with `g2p_en` fallback; ARPAbet phoneme input.
Telugu (new language)	Telugu Unicode block U+0C00–0C7F added to the symbol set; character-level pronunciation dictionary (one symbol per Unicode code point); an MFA acoustic model trained on the Telugu subset because no pretrained Telugu aligner exists; digits expanded to Telugu words; NFC normalisation; punctuation becomes a short-pause symbol.
Segmentation (final run)	Utterances are cut at silences ≥80 ms into 2–7 s segments using the existing MFA TextGrids (no re-alignment); segments inherit the split of their parent utterance.
New scripts	`inference.py` (frontend, synthesis, variance plots), `plot_losses.py` (TensorBoard curves), `subsample.py`, `segment.py`, `prepare_telugu_mfa.py`.

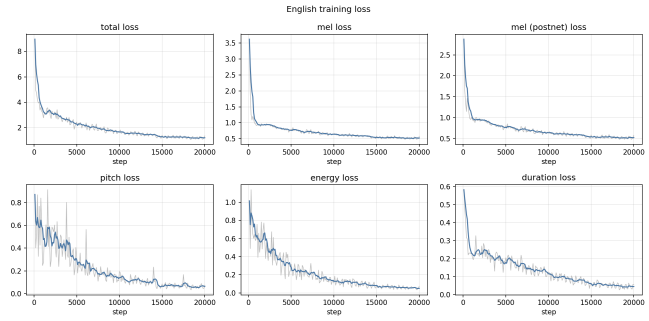
3. Dataset, preprocessing and training configuration

	English	Telugu
Source	Svarah (`ai4bharat/Svarah`, gated; test split only), Indian-accented English	IndicTTS Telugu (`SPRINGLab/IndicTTS_Telugu`), female voice only, 16 kHz upsampled to 22 050 Hz
Selection	Utterances 1–15 s; ~5.0 h train from 101 speaker groups, ~1.0 h validation from 14 other groups (whole speaker groups held out; groups inferred from a metadata fingerprint, 115 total)	Utterances 1.5–20 s, duplicate sentences removed, random split ~5.0 h / ~1.0 h, no sentence in both
After segmentation	3 909 utterances → 4 524 segments	1 853 utterances → 3 637 segments (median parent utterance 18 s)
Used for training	3 695 train / 751 validation segments (4.59 h in total after silence trimming and alignment drops)	3 030 train / 607 validation segments (3.92 h in total)
Alignment	MFA 3.4.3, pretrained `english_us_arpa`	MFA 3.4.3 trained on the subset, character dictionary

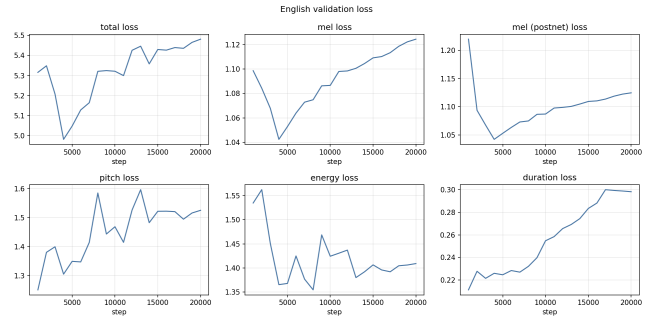
Training: 20 000 steps, batch size 16, gradient accumulation 1, Adam with warm-up 2 000 steps, learning-rate decay at steps 14 000 and 18 000, checkpoint and validation every 5 000 / 1 000 steps. One Tesla T4 per language, about 2–2.5 h each. Both runs were stopped by the notebook idle limit and resumed from the step-15 000 checkpoint. Checkpoints published: step 5 000 and step 20 000 (float16, model weights only, to stay under GitHub's size limit); step 20 000 is the official result.

Baseline run. A first run on whole utterances (30 000 steps) produced unintelligible audio with higher validation loss (English 4.92 at step 2 000 rising to 5.68; Telugu 3.88 rising to 4.48). Telugu clips of up to 20 s exceeded the model's sequence limit, which motivated the segmentation above. The segmented run is the one reported everywhere else.

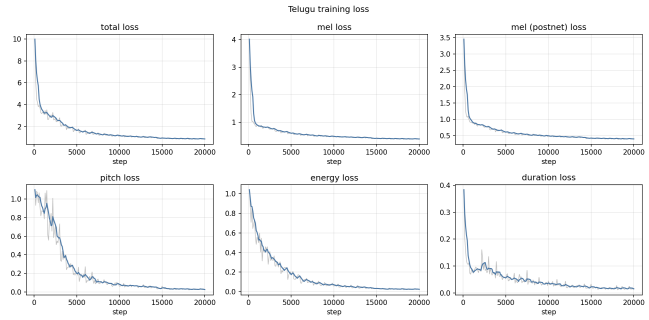
4. Loss curves



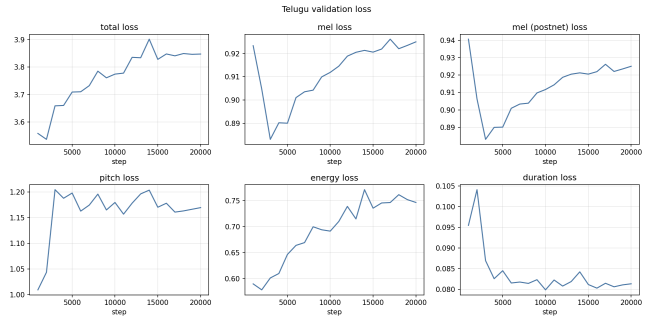
English training loss



English validation loss



Telugu training loss



Telugu validation loss

	Training total	**Validation total**	**Validation components (start → end)**
English	~9 → ~1.36 (mel ≈0.5, pitch/energy ≈0.05–0.07, duration ≈0.05)	5.32 (step 1 000) → **min 4.98 at step 4 000** → 5.48 at step 20 000	mel 1.04 → 1.12; pitch ~1.25–1.6; energy ~1.35–1.56; duration 0.21 → 0.30
Telugu	~10 → ~0.89 (mel ≈0.4, pitch/energy ≈0.03, duration ≈0.015)	3.56 (step 1 000) → **min 3.54 at step 2 000** → 3.85 at step 20 000	mel 0.88 → 0.925; pitch 1.0 → 1.17–1.2; energy 0.58 → 0.75; duration ~0.08

Analysis

- **Convergence.** Training loss falls smoothly and monotonically (the learning-rate decay at 14 000 / 18 000 gives a small extra drop) with no divergence or loss spikes. The resumed segment after step 15 000 joins the curve without a visible jump.
- **Overfitting.** Validation loss reaches its minimum very early (steps 2 000–4 000) and then rises while training loss keeps falling: the final train/validation gap is about 1.36 vs 5.48 (English) and 0.89 vs 3.85 (Telugu). The variance terms (pitch, energy) overfit most: training pitch/energy loss is 0.03–0.07 but validation stays 1.0–1.6, i.e. the model memorises the per-phoneme prosody of the training set instead of learning a general mapping. With about 4 h of audio, a ~35 M parameter model memorises quickly.
- **Instability.** The validation curve is noisy between evaluations (English more than Telugu: multi-speaker, varied recording conditions in Svarah), but there are no divergences.
- **Checkpoint choice.** The lowest-validation-loss checkpoint (steps 2 000–4 000) was not used as the official one because loss does not track intelligibility here; both step 5 000 and step 20 000 outputs are provided (`Results/<Language>_compare_step5000`).
- **Segmentation effect.** Compared with the whole-utterance baseline, the segmented run has lower validation loss and a smaller train/validation gap, but the audio is still mostly

unintelligible, so lack of data and the resulting memorisation, not only sequence length, is the main limit.

- **Vocoder sanity check.** Ground-truth mels from the training data passed through HiFi-GAN give clean speech, so artefacts come from the acoustic model's predicted mels.

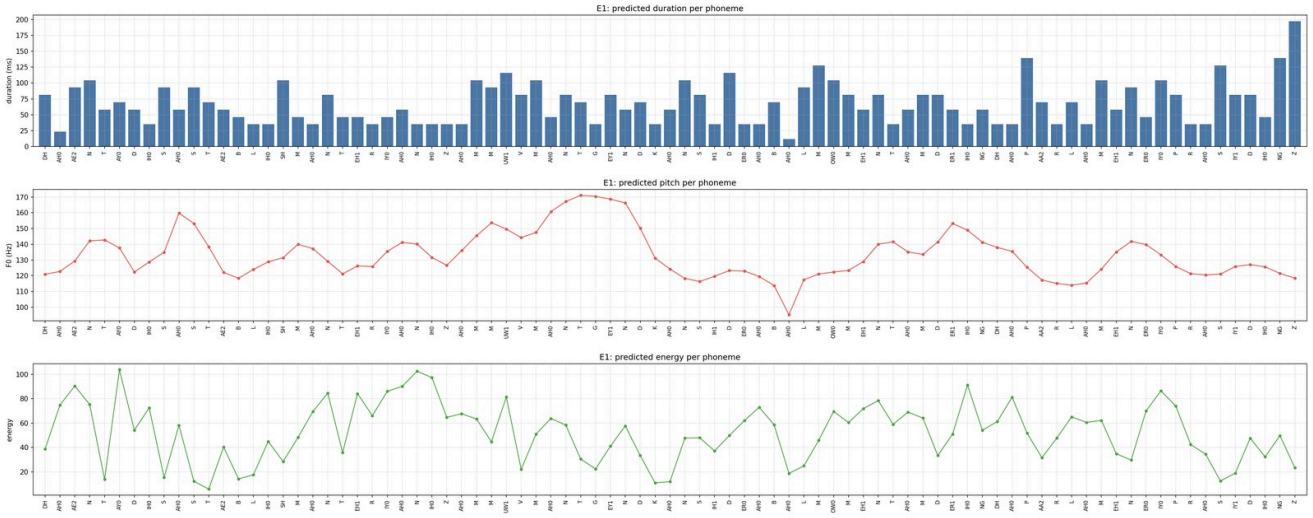
5. English test sentences (step 20 000, default speaker S073)

Subjective notes come from one listener (the author) and are not a formal MOS. Audio length: E1 5.77 s, E2a 4.53 s, E2b 3.92 s, E3 2.47 s, E4 6.87 s. Predicted F0 lies mostly between 95 and 170 Hz (male training speaker).

E1 – long word

> The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.

Speech: noisy, words not reliably understandable. The long word is out of lexicon, so it goes through `g2p_en`, giving a long phoneme string; the rest of the sentence uses lexicon pronunciations.

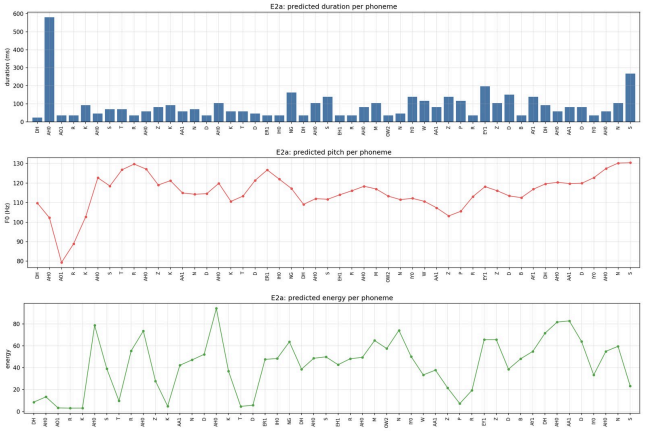


Duration: most phonemes are 35–105 ms, with phrase-final lengthening (~200 ms on the last Z of "proceedings") and longer vowels in "movement", "momentum" and "parliamentary"; 35 ms is the floor of the predicted values (a few frames), i.e. timing is coarse. **Pitch:** a broad rise to ~170 Hz on "movement gained" (the subject phrase), declination afterwards, a smaller rise in "momentum" and a flat tail; a realistic declarative shape. **Energy:** high at the onset ("The antidis-"), around "establishmentarianism" and "movement", lowest on unstressed schwas and stops. The variance plots therefore look sensible; the unintelligibility lies in the mel decoder, not in the predictors.

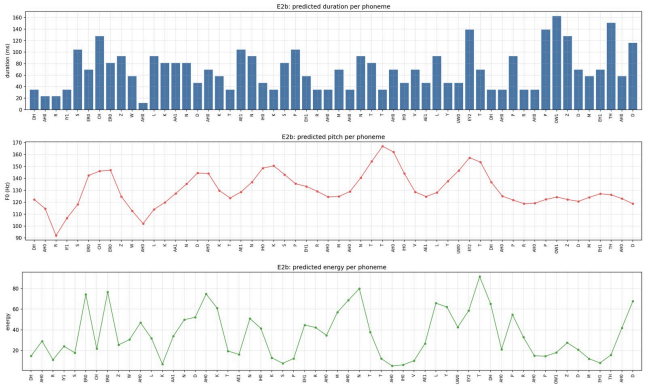
E2a / E2b – homograph "conduct"

> E2a: The orchestra's conduct during the ceremony was praised by the audience.
> E2b: The researchers will conduct an experiment to evaluate the proposed method.

Speech: both noisy and not understandable. **Frontend finding:** in both sentences "conduct" received the same pronunciation, `K AA1 N D AH0 K T` (noun stress). The verb reading in E2b should be `K AH0 N D AH1 K T`. The frontend (CMU lexicon lookup, no part-of-speech disambiguation for this word) therefore fails the homograph test; this is a frontend limitation, independent of model quality.



E2a



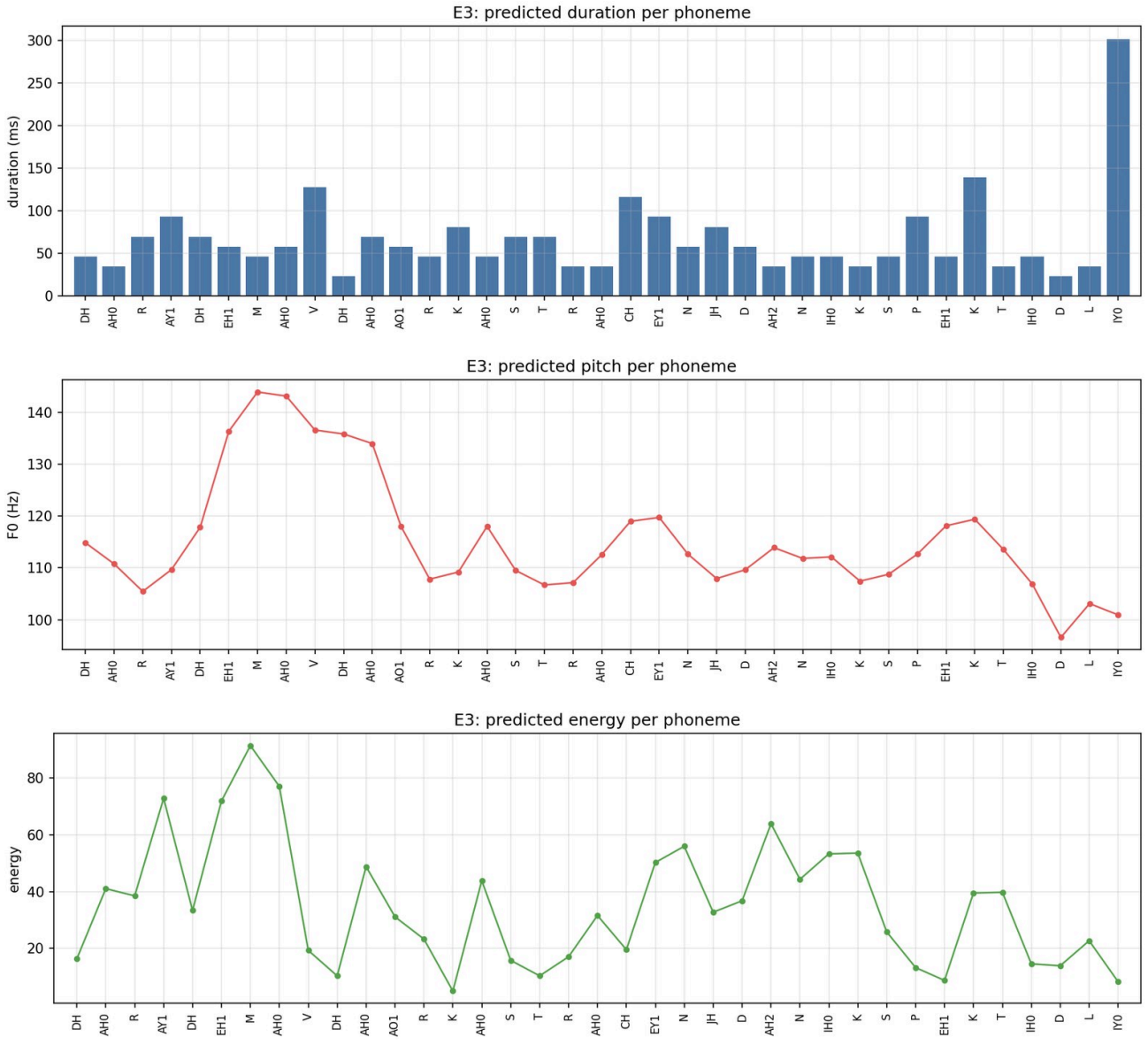
E2b

E2a: the first phoneme sequence ("The orchestra's") contains a ~580 ms prediction for the schwa after DH, an outlier caused by the model's confusion at utterance start; the sentence-final S is long (~270 ms). Pitch starts very low (~80 Hz at AO1) and climbs to ~130 Hz at the end, i.e. a rising rather than falling contour, which sounds unnatural for a statement. Energy is near zero over "DH AH0 AO1 R K" and spikes on AH0 in "orchestra", the energy predictor confusing the onset. **E2b:** durations are well-behaved (35–160 ms, longest on OW1 in "proposed" and TH in "method"); pitch rises on "researchers", at the end of "experiment" and on "evaluate" (peaks 145–167 Hz) and then flattens to ~120 Hz for "the proposed method". Energy dips to ~5 on unstressed function words. Thus the predictors give sentence-appropriate but not noun/verb-specific prosody for "conduct".

E3 – "rhyth m"

> The rhyth m of the orchestra changed unexpectedly.

Speech: unintelligible. The text is kept exactly as given, including the space inside "rhyth m"; the frontend therefore treats it as two words: "rhyth" → `R AY1 DH` and "m" → `EH1 M`, not `R IH1 DH AH0 M`. The split pronunciation is a faithful consequence of the input and is what the plot shows.

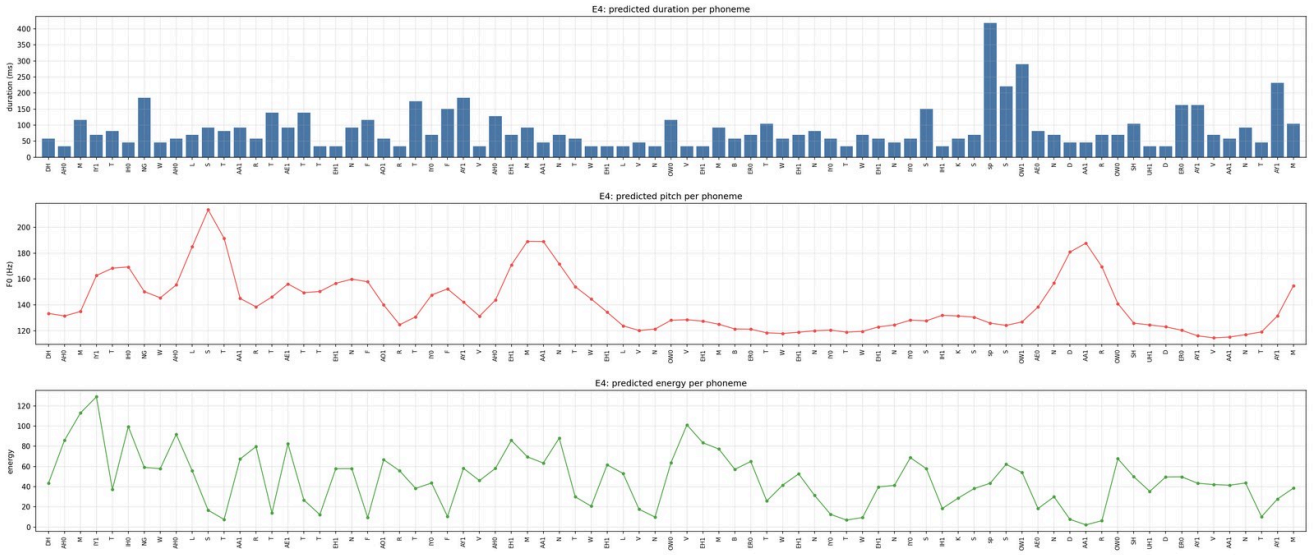


Duration: mostly 35–115 ms; the final IY of "unexpectedly" is ~300 ms (phrase-final lengthening, somewhat too long). **Pitch:** a strong peak (~144 Hz) at the split "EH1 M AH0 V" region, then a flat ~105–120 Hz line and a fall to ~97 Hz at the end. **Energy:** highest (~90) on M of the split word, so the artificial word break creates a prosodic "emphasis" that natural text would not have.

E4 – numbers and a non-dictionary word

> The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

Speech: unintelligible. The normaliser expands "10:45 AM" to "ten forty-five A M", "12" to "twelve", "2026" to "twenty twenty-six"; "andaroo" is out of lexicon and is spelled by `g2p_en` as `AE0 N D AA1 R OW0`.



Duration: the comma becomes an `sp` token with the longest prediction in the sentence (~420 ms) followed by long "S OW1" (~290 ms) at the start of "so": the model predicts a clear clause boundary there, which matches the text. Phrase-final M is ~100 ms, shorter than in E1. **Pitch:** peaks at the start of the date phrase ("start at ...", ~215 Hz), at "A M on" (~190 Hz) and again after the pause on "andaroo" (~188 Hz), flat ~120 Hz in between; the shape follows the three information-bearing groups. **Energy:** very high at the very beginning (up to ~130, an onset overshoot), moderate afterwards, near zero at the pause.

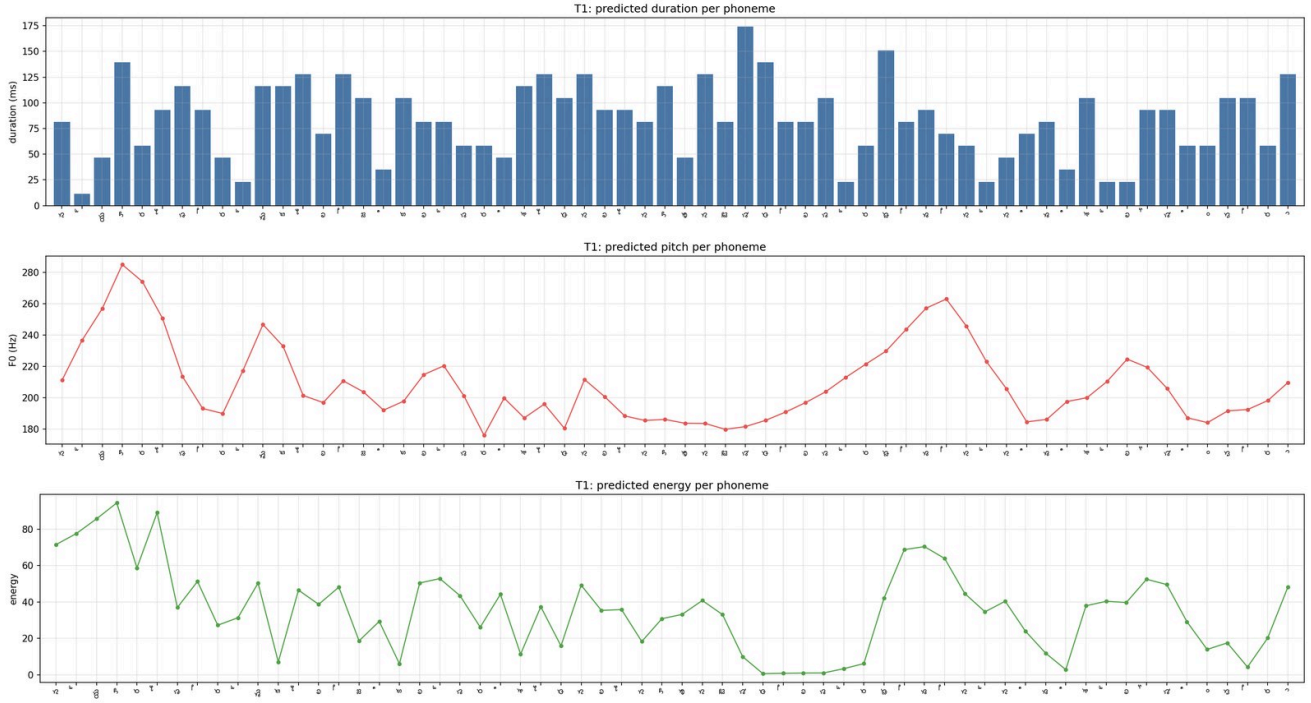
6. Telugu test sentences (step 20 000, single female speaker)

Audio length: T1 5.17 s, T2a 2.61 s, T2b 2.88 s, T3 2.16 s, T4 8.10 s. Predicted F0 is mostly 160–310 Hz (female speaker). x-axis labels are single Unicode code points (the unit of the model), so vowel signs appear as separate marks.

T1 – long technical word

> న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

Speech: unintelligible. The long loan word with many conjuncts (virama sequences) is far from the training vocabulary.

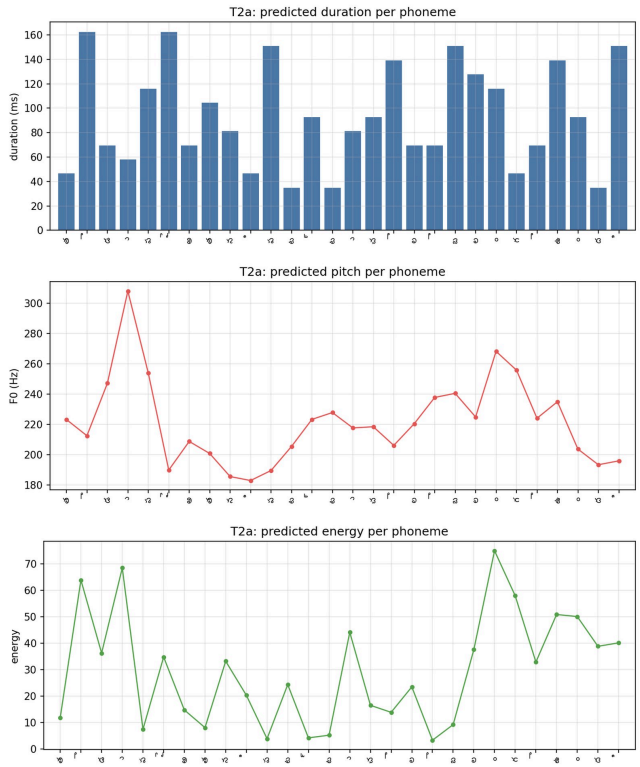


Duration: 10–175 ms; a few very short values (~10–25 ms) on virama/vowel-sign symbols (they carry no sound of their own), which is expected for a code-point inventory. **Pitch:** starts high (~285 Hz) on the first syllables, descends to ~180 Hz mid-sentence and has a second, smaller peak (~263 Hz) at "ప్రభావాన్ని"; consistent with the declining-with-emphasis pattern of the training speaker. **Energy:** high at the onset (~95), a long nearly silent stretch (~0–10) around the middle where the predictor expects a pause between words, and a second bump at the same peak as pitch.

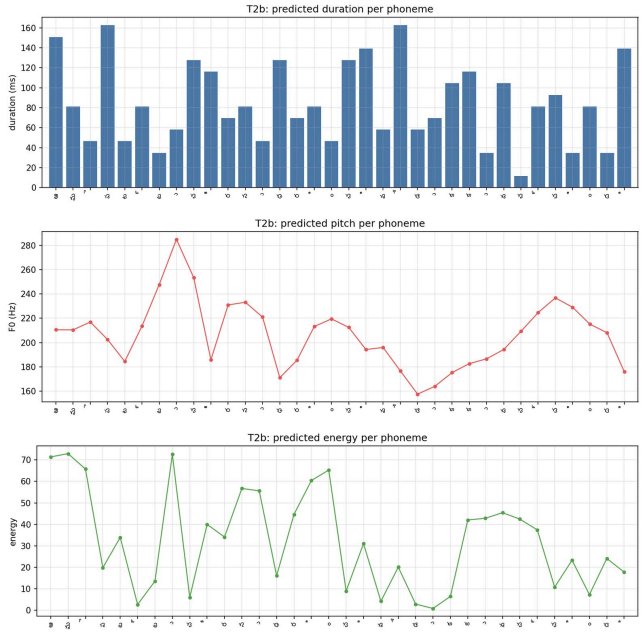
T2a / T2b – same spelling "పట్టు", different meaning

> T2a: తాడుపై అతని పట్టు చాలా బలంగా ఉంది.
> T2b: ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.

Speech: the first ~2 s of both clips are somewhat understandable; the rest is not. "పట్టు" ("grip" in T2a, "silk" in T2b) has the same spelling and the same pronunciation, so a text-only, code-point frontend gives identical phoneme symbols for it. The sense difference can influence prosody only through context.



T2a



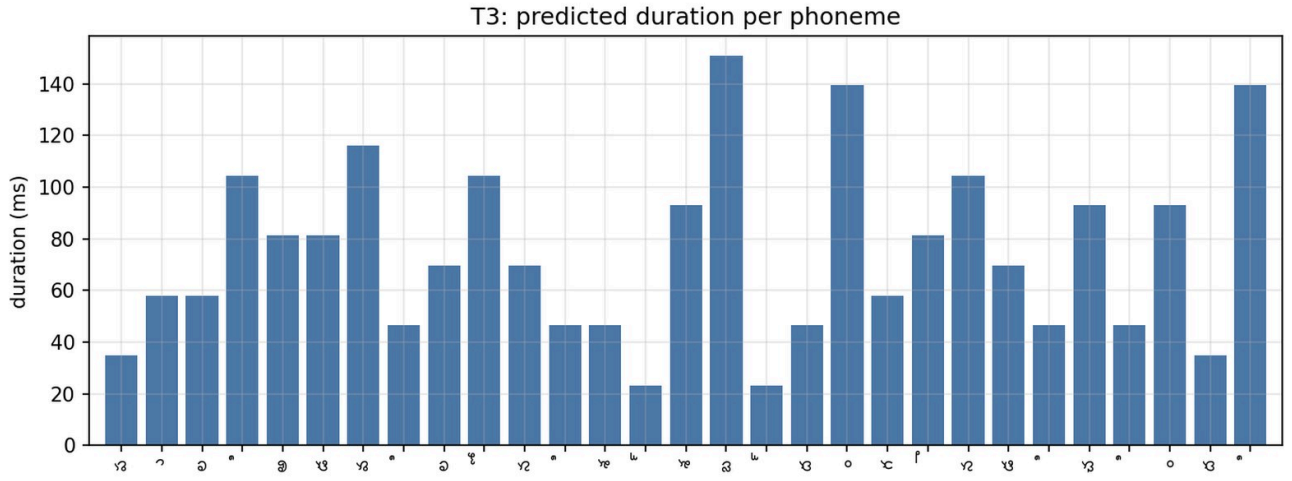
T2b

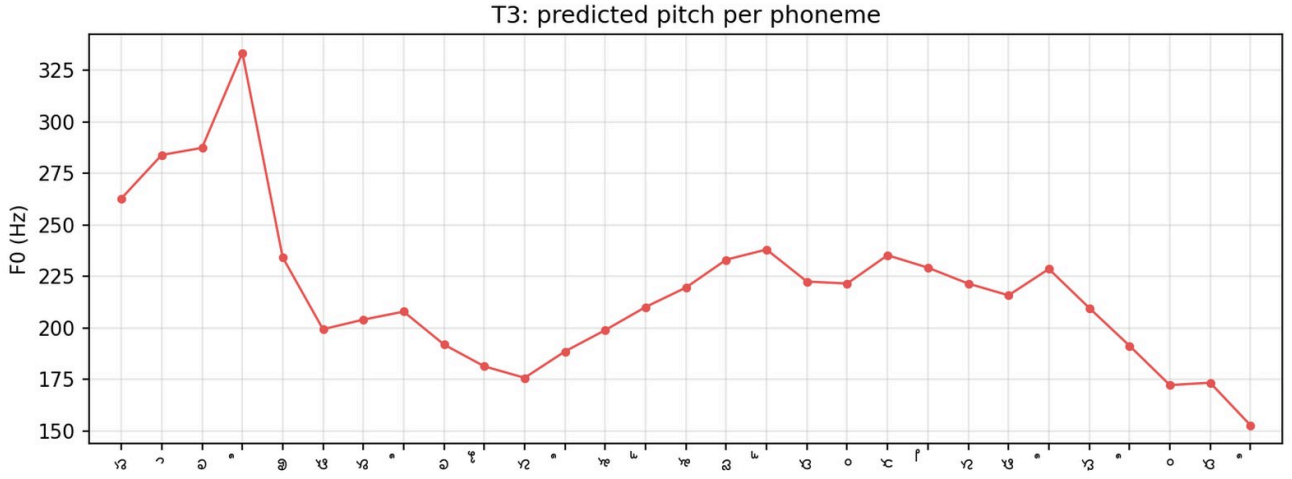
T2a: durations 35–165 ms; pitch spike at 307 Hz early (in "తరుచై" ~ the first stressed syllable), low trough ~183 Hz mid-sentence, second rise to ~268 Hz near "చాలా". Energy bursts at the start and again in the second half ("బలంగా ఉంది "). **T2b:** pitch peak ~285 Hz on the first stressed syllable of "చిరను" region; lowest ~157 Hz in the second half ("ధరించి") with a final fall to ~176 Hz; energy is highest in the first word and drops to ~0–10 for the middle section. So the predictors give different contours for the two sentences from sentence context only; the homograph itself is not resolved by sense (it cannot be).

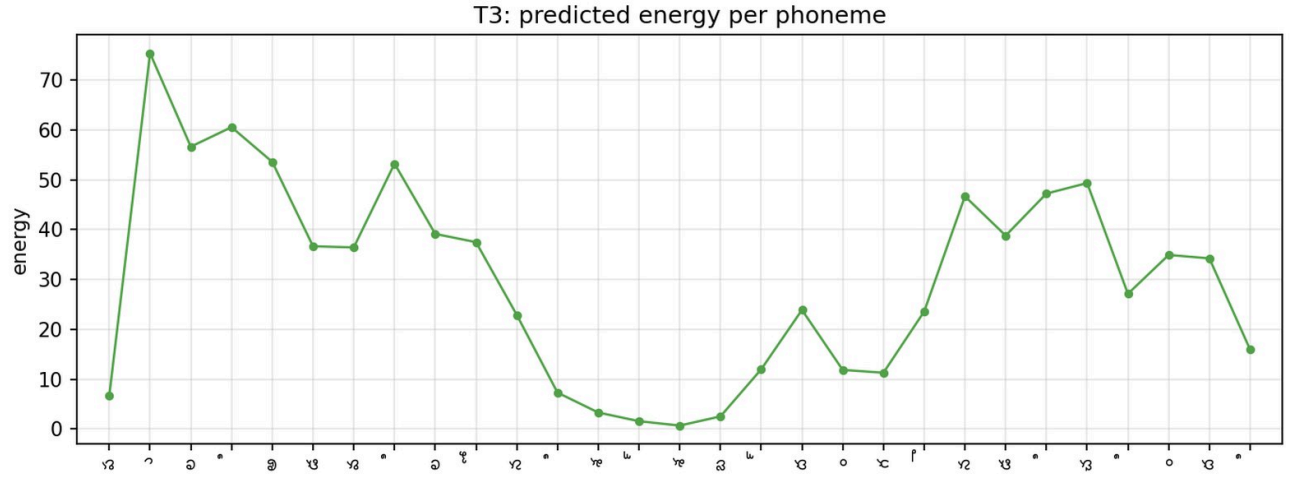
T3 – odd spacing

> పులి అడ విలో నిశ్శబ్దంగా నడిచింది.

Speech: unintelligible. The input is kept as given: "అడ వీలో" contains a split word (“అడ వీలో” written as "అడ" + double space + "వీలో"), so the frontend sees two tokens. Word boundaries are not given their own symbols in this model; the two spaces make no extra pause token, so the effect shows only through the sequence itself.





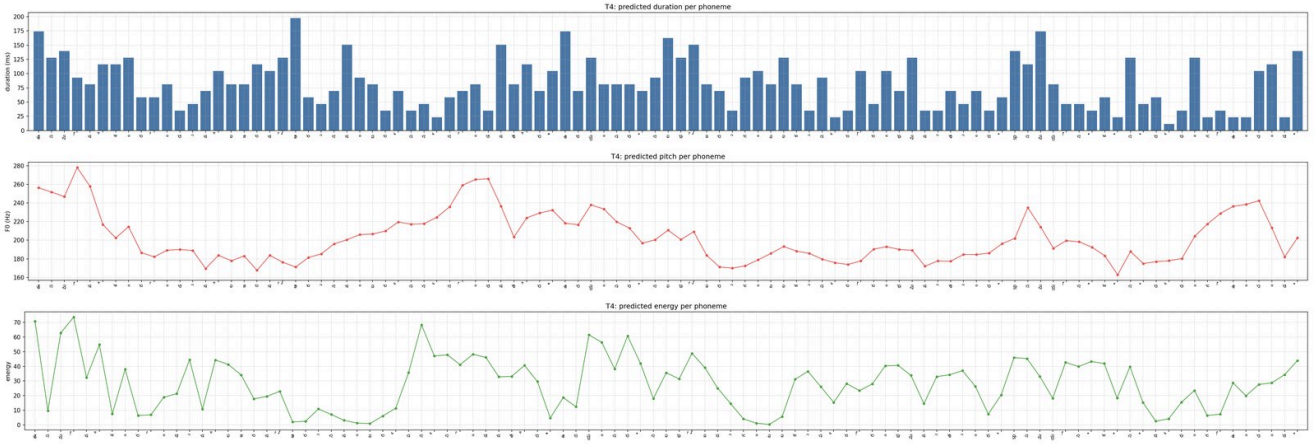


Duration: narrow range (23–150 ms), no pause symbol; the longest values come on "అడ వీలో" and the final syllable ("...ుంది ", ~140 ms). **Pitch:** very high start (up to ~333 Hz at the 4th symbol), then a slow decline to 152 Hz: an exaggerated opening that does not match typical declination. **Energy:** strong in the first 4 symbols, collapses to ~1 around the split "అడ వీలో" (the energy predictor treats the gap as silence), then recovers for the final word.

T4 – numbers, date and comma

> ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

Speech: unintelligible. Numbers are expanded into Telugu words before synthesis (2026 → రెండు వేల ఇరవై ఆరు, 12వ → పన్నెండవ, 10:45 → పది నలభై ఐదు), then NFC-normalised; the expansion adds many symbols, the longest input of the Telugu set.



Duration: uniformly 25–200 ms; the comma becomes an `sp` token of ~140 ms (shorter than the ~420 ms English comma), which is a weak clause boundary. **Pitch:** repeated descents from ~265 Hz to ~170 Hz over the number words, a small rise (~235 Hz) at the clause boundary after the comma and a final rise to ~243 Hz at the end, which makes the last clause sound like a question or continuation. **Energy:** alternates between ~70 (word onsets) and near zero (long gaps inside the expanded number words), with the longest low-energy valley inside "…ఇరవై ఆరు" — the predictor sees these as pause-like.

7. English vs Telugu

	English	Telugu
Data	multi-speaker (115 speaker groups), accented, varied recording conditions; ASR benchmark, not TTS data	single studio-quality speaker (TTS corpus), 16 kHz upsampled
Validation loss (final)	5.48 (best 4.98 at step 4 000)	3.85 (best 3.54 at step 2 000)
Train/validation gap	1.36 vs 5.48	0.89 vs 3.85
Frontend	g2p / lexicon: homograph and out-of-lexicon failures (E2, E4)	character-level: no dictionary needed, but no schwa/sandhi rules and no way to treat homographs
Intelligibility	noisy to unintelligible	first ~2 s of T2a/T2b understandable; others unintelligible

Why Telugu loss is lower yet audio is not better: a single speaker and a clean studio voice make the mel targets easier to regress, so lower numbers do not mean better speech; the absolute values are also not comparable across languages. The English model has to explain 115 voices with 3 695 segments, which is a much harder problem at this data size.

Limitations and failures

- ~4 h of audio per language is far below what FastSpeech 2 needs (the reference uses ~100 h+); the models memorise the training set.

- Svarah is an ASR corpus with noisy, spontaneous speech and uneven recording quality; speaker groups are inferred from metadata.
- Mel decoder output is blurry on unseen inputs; the vocoder then amplifies the artefacts as noise.
- Frontend gaps: homograph handling (E2b), mispronounced split words (E3, T3), no Telugu lexicon rules.
- MFA alignment quality for Telugu (aligner trained on a small subset) bounds the duration targets; MFA 3.4.3 crashed after training for single-speaker corpora and had to be worked around (see README).
- Evaluation is a single-listener judgement; no objective metric (MCD, WER from an ASR) was computed.

Possible improvements

- More data (the full IndicTTS Telugu, a proper TTS corpus for English such as LJSpeech/LibriTTS subsets) or fine-tuning from a pretrained FastSpeech 2.
- Stronger regularisation, early stopping at the validation minimum, data augmentation (speed/pitch perturbation) and smaller model size.
- Frontend: POS-aware homograph resolution; a Telugu grapheme-to-phoneme module with proper conjunct and schwa handling.
- Fine-tune the HiFi-GAN vocoder on the target speakers; add pauses at word boundaries; ASR-based intelligibility scoring.

8. Reproducibility

- Repository layout, `requirements.txt` (exact versions used: torch 2.11.0, librosa 0.11.0, g2p_en 2.1.0, etc.), all configs in `English/config_dir` and `Telugu/config_dir` (merged `config.yaml` for reading) and the exact data split (`Report/data_manifests`, `tools/segment_manifests`).
- Seed: the split is fixed by the manifest files; training seeds are those of the reference code.
- Commands: see README – `python subsample.py` → MFA alignment → `tools/segment.py` → `tools/setup_seg.py` → `python preprocess.py config_dir/preprocess.yaml` → `python train.py -p ... -m ... -t ...` (resume with `--restore_step 15000`) → `python inference.py --restore_step 20000 ...` → `plot_losses.py`.
- Environment: Kaggle notebook, Python 3.13, Tesla T4; Montreal Forced Aligner 3.4.3 installed separately.
- Checkpoints: steps 5 000 and 20 000, float16 model weights only (<100 MB each).
- Test sentences are fixed in `inference.py` and kept exactly as specified (including spacing in E3 and T3).